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Theme A FinSwarm

For TC1

FINAL CEO DECISION

As CEO of FinNova Capital, I have synthesized the inputs from our Research, Finance, Credit, and Compliance departments. We are moving forward with the **"Optimized Blend" Pilot Strategy**, prioritizing capital efficiency and risk-adjusted returns.

Executive Decision: The INR 27 Cr Lending Pilot

1. Customer Segment Mix

We will focus exclusively on two segments to maximize capital velocity while maintaining a conservative risk profile:

- * **Service SMEs (70% of capital):** 315 loans at INR 6L average.
- * **Manufacturers (30% of capital):** 90 loans at INR 9L average.
- * **Exclusion:** Retail Shops are excluded due to high acquisition costs and unfavorable risk-to-loan-size ratios.

2. Product Terms

- * **Interest Rates:** 18% (Service SMEs) / 19% (Manufacturers).
- * **Loan Structure:** "Cash-Flow First" underwriting. 80% of max limit for initial disbursement, with a "top-up" facility for borrowers demonstrating perfect repayment behavior over 6 months.

3. Approval Policy

- * **Underwriting:** Prioritize GST filing consistency and digital payment receipts.
- * **Compliance:** Zero-tolerance for predatory practices. All APRs, fees, and penalty clauses must be disclosed in plain language. No hidden administrative costs.

4. Budget Allocation

- * **Total Acquisition Budget:** INR 42 Lakh.

- * **Initial Deployment:** INR 15.97 Lakh for the 405-loan acquisition.
- * **Contingency Reserve:** The remaining INR 26 Lakh is frozen as a "Credit Loss Contingency Fund," accessible only upon successful completion of Phase 1 and audit of collections infrastructure.

5. Risk Limits

- * **Portfolio Default Ceiling:** Hard cap at 5%.
- * **Concentration Limit:** No segment to exceed 70% of total capital.
- * **Treasury Guardrail:** If quarterly default rates exceed 4%, an automatic 50-100 bps interest rate hike will be triggered for new originations.

6. Go-To-Market (GTM) Approach

- * **Channel Strategy:** Focus on high-intent digital channels (Search/B2B platforms) to ensure low CPA.
- * **Phased Rollout:**
 - * **Phase 1 (Months 1-3):** Deploy 30% of capital to Service SMEs to stress-test collections.
 - * **Phase 2 (Months 4-9):** Scale to full deployment.
 - * **Phase 3 (Months 10-12):** Halt new originations; focus on collections and data gathering.

Key Performance Indicators (KPIs)

Metric	Target
Net Interest Margin (NIM)	Maintain a floor of 3.0% post-default and post-fixed costs.
Portfolio Default Rate	Keep below 3.83% (Weighted Average) with a hard stop at 5.0%.
Capital Velocity	100% deployment of the INR 27 Cr within the 9-month acquisition window.

Implementation Sequence

Treasury will immediately establish the escrow accounts.

The Risk Committee is directed to finalize the "Cash-Flow First" underwriting manual by EOW for final CEO sign-off.

Marketing is authorized to initiate Phase 1 lead generation effective immediately.

For TC2

FINAL CEO DECISION

As CEO of FinNova Capital, I have synthesized the inputs from Finance, Risk, Compliance, and Marketing to finalize the **Pilot Redesign Strategy**. We are shifting from a high-growth, high-risk posture to a **sustainable, risk-adjusted profitability model**.

Executive Decision: The "Stability-First" Pilot Pivot

We are immediately implementing a portfolio rebalancing and tightening of underwriting standards to align with our 5.5% EDR mandate.

- * **Customer Segment Mix:** Shift to a defensive posture: **55% Service SMEs** (Anchor), **25% Retail**, and **20% Manufacturing**.

- * **Product Terms:** Interest rates set at **18.5%** for Retail/Manufacturing and **17%** for Service SMEs.

- * **Approval Policy:** Hard rejection for CIBIL <700. Implementation of "Additional Business Evidence" (GST/Bank Statements) for all applicants. Tiered underwriting: Tier 1 (Automated) for >720; Tier 2 (Bank statements) for 650–719; Tier 3 (Manual/Full documentation) for <650.

- * **Budget Allocation:** Total deployment of INR 24.3 Crore (INR 27 Cr minus 10% Liquidity Buffer). Marketing spend is re-allocated to high-intent channels (SEM/SEO) to prioritize quality over volume, maintaining a strict CPQA ceiling.

- * **Risk Limits:** 15% safety margin on internal scoring. Automatic "Credit Freeze" triggered if any segment exceeds 85% of its default ceiling.

- * **Go-To-Market Approach:** Pivot from broad-reach to hyper-targeted, industry-specific lead aggregation to ensure the Approval-to-Funded ratio remains healthy under tighter screening.

- * **Implementation Sequence:**

- * **Days 1–15:** System integration of enhanced underwriting and automated rejection protocols.

- * **Days 16–30:** Full deployment of revised pricing and segment-specific marketing.

- * **Ongoing:** Daily monitoring of Approval-to-Disbursement ratios; weekly Treasury audits.

Key Performance Indicators (KPIs)

1. **Portfolio Expected Default Rate (EDR):** Maintain a rolling 30-day EDR at or below **5.4%** (Targeting the 5.5% mandate with a 0.1% safety margin).
2. **Net Interest Margin (NIM):** Achieve a sustained **6.5%** NIM through optimized pricing and cost-of-funds discipline.
3. **Approval-to-Disbursement Ratio:** Maintain a ratio between **65%–70%**. (If the ratio drops below 60%, it indicates over-tightening; if it exceeds 75%, it indicates a breach of risk controls).

Compliance & Operational Mandate

All departments are directed to operate within existing headcount and infrastructure. No additional external spend is authorized. The 10% Liquidity Buffer (INR 2.7 Crore) is ring-fenced and non-negotiable. We are prioritizing long-term institutional viability over short-term volume.

Decision Status: **Effective Immediately.**

For TC3

FINAL CEO DECISION

As CEO of FinNova Capital, I have synthesized the departmental inputs into the following executive decision. We are pivoting from a volume-based growth strategy to a **Quality-First Portfolio Model**.

We formally reject the 160-loan target as a vanity metric that risks our capital adequacy and credit integrity. Our new objective is the disciplined origination of **110** high-quality, profitable small-business loans.

1. Strategic Parameters

- * **Customer Segment Mix:** Focus exclusively on "Accountant-Verified" leads (Tier 1 & 2). We will deprioritize broad digital acquisition to avoid low-intent, high-default profiles.
- * **Product Terms:** Standardized, transparent term loans. No balloon payments. APRs will be risk-adjusted based on the Tier 1–3 credit scoring model.
- * **Approval Policy:** Strict adherence to the 40% DTI ceiling (35% for Tier 3). Any application flagged as "High Risk" by our fraud engine is subject to immediate, non-appealable decline.

- * **Budget Allocation (INR 18 Lakh Total):**
 - * **Referrals:** INR 1.44 Lakh (Maximized for high ROI).
 - * **Accountants:** INR 11.7 Lakh (Primary channel; acts as the "Trust Anchor" and initial KYC filter).
 - * **Digital Ads:** INR 4.86 Lakh (Performance-gated; will be cut if default rates exceed 5% in the first 60 days).
- * **Risk Limits:** 15% liquidity reserve on all deployed capital. No single industry/geography to exceed 10% of the total loan book.
- * **Go-to-Market Approach:** "Trust-Centric" acquisition. We will launch a Transparency Portal to ensure full disclosure of fees and APRs, reducing bounce rates and ensuring regulatory compliance (UDAAP/TILA).
- * **Implementation Sequence:**
 - * **Phase 1 (Month 1):** 30 loans (Testing underwriting efficiency).
 - * **Phase 2 (Month 2):** 40 loans (Scaling based on performance).
 - * **Phase 3 (Month 3):** 40 loans (Full deployment).

2. Key Performance Indicators (KPIs)

To ensure portfolio health and operational success, we will track the following:

1. **Portfolio Default Rate (PDR):** Must remain below the 2.5% Maximum Portfolio Default Rate (MPDR) threshold.
2. **Cost Per Funded Loan (CPFL):** Blended target of <INR 7,000 across all channels, with a specific focus on maintaining the Accountant channel efficiency.
3. **Early Warning System (EWS) Trigger Rate:** Percentage of portfolio showing behavioral deterioration (e.g., missed utility payments, credit score drops) must be <2% monthly to avoid immediate pause in originations.

3. Executive Directive

The Treasury desk is authorized to release funds in tranches.

Marketing is to pivot immediately to the Accountant-led strategy.

Compliance will perform a weekly audit of the Transparency Portal to ensure we remain above reproach regarding consumer protection standards.

This plan is approved for immediate execution.

For TC4

FINAL CEO DECISION

As CEO of FinNova Capital, I have synthesized the operational, financial, and strategic inputs to address our current verification bottleneck and regulatory mandate.

Executive Decision: The "Hybrid Scaled" Operational Pivot

To maintain our 48-hour SLA, ensure 100% regulatory compliance, and preserve liquidity, we are adopting the **Hybrid Scaled Model** effective immediately.

- * **Customer Segment Mix:** Focus on prime and near-prime applicants to ensure high-intent, lower-risk profiles that align with our new 40% manual verification requirement.
- * **Product Terms:** Standardized personal loan products with clear APR disclosures; no changes to existing interest rates to maintain competitive positioning.
- * **Approval Policy:** Shift to a 60/40 (Automated/Manual) verification split. All applications must pass automated KYC/KYB checks; the 40% manual review will focus on high-risk flags and complex income verification to ensure fair lending compliance.
- * **Budget Allocation:** Total investment of INR 10.7 Lakh. INR 8 Lakh (CapEx) for automated integration; INR 2.7 Lakh (OpEx) for temporary staffing. The remaining INR 4.3 Lakh is reserved as a liquidity buffer.
- * **Risk Limits:** Implementation of "Hard-Stop" underwriting rules: DTI ratio capped at 40%; automatic decline for applicants failing automated identity verification.
- * **Go-To-Market (GTM) Approach:** Shift marketing spend toward high-intent, organic, and referral channels to ensure lead quality, reducing the burden on the verification team.
- * **Implementation Sequence:**
 - * **Phase 1 (Weeks 1-2):** Onboard 2 temporary reviewers to clear the backlog and manage the 40% manual requirement.
 - * **Phase 2 (Weeks 1-2):** Initiate Automated Verification Service integration (milestone-based payments).
 - * **Phase 3 (Week 3+):** Transition to automated workflows; phase out temporary staff as efficiency gains are realized.

Key Performance Indicators (KPIs)

1. **Verification Throughput Efficiency:** Maintain a median approval time of **≤ 48 hours** for 95% of applications.
2. **Operational Cost-Per-Verification:** Achieve a **15% reduction** in manual verification costs by the end of Month 3 through the automated integration.
3. **Regulatory Compliance & Complaint Rate:** Maintain a customer complaint rate of **$< 2\%$** and ensure **zero** regulatory findings regarding verification failures during the transition.

Final Directive: The Finance department is authorized to release the initial 30% of the CapEx budget immediately.

All department heads are to align their internal workflows with this 48-hour SLA mandate.

We are moving from a manual-heavy culture to a tech-enabled, scalable infrastructure.

Proceed with execution.

For TC5(live testcase)

FINAL CEO DECISION

As CEO of FinNova Capital, I have synthesized the departmental inputs to address our current portfolio challenges. We are moving from a reactive stance to a **"Controlled Deployment"** model to preserve capital and ensure long-term viability amidst rising costs and fraud risks.

Executive Decision: Portfolio Re-optimization

- * **Customer Segment Mix:** 40% Retail (High-Yield/High-Touch) / 60% Non-Retail (Low-Risk/Institutional).
- * **Product Terms:** Retail pricing set at **19% APR** to offset the 13% Cost of Funds (CoF) and fraud-related leakage. Non-Retail yield target remains at 17.5%.
- * **Approval Policy:** "Screen-First" mandate. All retail applications must pass automated fraud screening (Device Fingerprinting, Velocity Checks, and Biometric verification). Manual underwriting is mandatory for all high-value or borderline cases.
- * **Budget Allocation:** Total deployable capital is **INR 21 Cr** (after maintaining the INR 3 Cr liquidity buffer). Retail: INR 8.4 Cr; Non-Retail: INR 12.6 Cr.

* **Risk Limits:** Aggregate portfolio default ceiling is **5.5%**. A "Management Buffer" of 15% below this ceiling is established; hitting this buffer triggers an automatic tightening of credit scoring models.

* **Go-To-Market Approach:** Shift marketing focus to high-intent, low-funnel channels to ensure lead quality. Implement "Credit-Gate" filtering at the application stage to prevent wasted acquisition costs.

* **Implementation Sequence:**

1. Immediate integration of the Fraud Screening API (Priority 1).
2. Update internal risk models to reflect 13% CoF and new fraud parameters.
3. Rebalance portfolio allocation to the 40/60 split.
4. 30-day post-launch audit to verify fraud reduction targets.

* **Measurable Outcomes:** Restoration of a 3.6% net interest spread and stabilization of the default rate below the 5.5% threshold.

Key Performance Indicators (KPIs)

1. **Fraud Incidence Rate:** Must be reduced from 7% to **$\leq 2.8\%$** within 30 days of implementing the "Screen-First" policy.
2. **Net Interest Margin (NIM):** Maintain a minimum spread of **3.6%** (Weighted Yield of 18.1% vs. 14.5% break-even cost).
3. **Portfolio Default Rate:** Must remain strictly below the **5.5%** ceiling; any breach triggers an immediate freeze on new retail disbursements.

Compliance Note: All marketing and underwriting activities must adhere to the "Fair Lending" mandate. Risk-based pricing at 19% must be transparently disclosed to all retail customers, and no predatory collection or rollover practices will be permitted.

This decision is effective immediately.

Tech stack used

- **Orchestration & State Management:** LangGraph (built on top of LangChain). Used to construct a stateful graph containing discrete agent nodes and cyclic review loops instead of rigid, linear prompt chains.
- **Core Language Model:** Google Gemini (gemini-3.1-flash-lite) accessed via the Google Generative AI SDK. Selected specifically for its ultra-low latency, multimodal comprehension, and high throughput on developer-tier rate limits.
- **Environment & Runtime:** Python 3.10+, managed locally using python-dotenv for configuration safety and isolated environment variables.

How It Was Structured & Envisioned

1. The Core Philosophy (Why Multi-Agent?):

A single monolithic prompt trying to act as an entire corporate boardroom (balancing risk, finance, compliance, and growth simultaneously) typically results in superficial answers or hallucinations. The system was envisioned as a digital simulation of real corporate friction: forcing distinct personas with opposing metrics (e.g., Finance wanting low risk vs. Marketing wanting broad deployment) to argue through an iterative cycle.

2. State Machine Over Chatbots:

Instead of a chat interface where a model free-associates, the architecture treats the boardroom like a finite state machine. A typed dictionary (BoardroomState) acts as the shared ledger passing intermediate findings from node to node:

**Business Brief → Research → Finance → Credit Risk → Compliance
→ Marketing → Increment Cycle ↻ → CEO Decision**

3. Engineering Around Free-Tier Reality (The Fail-Safes):

Building this realistically meant designing for failure. Free-tier API keys hit rate limits instantly when 5 to 8 sequential LLM calls fire back-to-back. To solve this without spending enterprise budgets:

- A multi-key rotation wrapper (invoke_with_key_switching) catches rate-limit exceptions and hot-swaps to a secondary API key on the fly.

- Intentional pacing (`time.sleep`) was added to nodes to throttle requests safely within platform thresholds.

4. Iterative Prompt Hardening:

The agents started too agreeable—they would praise each other's ideas. Building a high-scoring simulation required injecting adversarial system instructions: explicitly programming the Credit Risk and Finance agents to aggressively challenge high-exposure strategies, while anchoring the CEO to strict constraints like capital reserves, default ceilings ($\leq 5\%$ or 5.5%), and interest rate caps ($\leq 19\%$).